

Cybersecurity Basics for Beginners

by OpenLibrary Sample Team

A friendly first look at how attacks work and how to stay safe online.

What is cybersecurity?

Cybersecurity means protecting computers, networks and data from people who should not have access to them.

Security experts describe the goal with three words: confidentiality, integrity and availability. Confidentiality means only the right people can see the data. Integrity means the data is not changed by mistake or by an attacker. Availability means the system works when you need it.

Every security decision is a trade-off between these three goals and how easy the system is to use.

Common threats

Phishing is a fake message that tricks you into giving a password or clicking a bad link. Malware is software written to cause harm, such as viruses, spyware and ransomware. Ransomware locks your files and asks for money to unlock them.

Most attacks do not begin with clever hacking. They begin with a person who trusted a message too quickly. Slow down when a message creates fear or urgency.

Simple habits that protect you

Use a different long password for every account and keep them in a password manager.

Turn on two-factor authentication wherever it is offered. Install updates soon, because updates fix known security holes. Back up important files in a second place.

No single habit makes you safe, but together they stop most everyday attacks.